Cognitive Dysfunction Syndrome
A Disease of Canine and Feline Brain Aging

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KEYWORDS

• Cognitive dysfunction syndrome • Brain aging • Behavior • Canine • Feline

KEY POINTS

- Brain aging is a degenerative process that for many dogs and cats ultimately progresses to a loss of one or more cognitive domains or impairment of cognitive function.
- Diagnosis of cognitive dysfunction syndrome (CDS) is based on recognition of behavioral signs and exclusion of other medical conditions and drug side effects, which in some cases can mimic or complicate CDS.
- Clinical categories include disorientation, alterations in social interactions, sleep-wake cycles, elimination habits, and activity, as well as increasing anxiety. Deficits in learning and memory have also been well documented.
- Treatment is aimed at slowing the advancement of neuronal damage and cell death and improving clinical signs. Drugs, diet, and supplements can be used alone or concurrently to improve neurotransmission and reduce oxidative damage and inflammation.

INTRODUCTION

As pets age, behavior changes may be the first indication of declining health and welfare. This is particularly true for some of the most common problems associated with aging, such as pain, sensory decline, and cognitive dysfunction syndrome (CDS). Early identification of these signs provides an opportunity for effective intervention.

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During veterinary visits, pet owners are likely to report serious behavioral changes, but subtle signs, which may be indicative of declining health or cognition, often go unreported. Family members therefore need assistance in both identifying and reporting any change from normal behavior to their veterinarian. Similarly, clinicians must be proactive in asking about behavioral signs.

The current article focuses on how CDS in dogs and cats parallels neurodegenerative disorders in humans, particularly Alzheimer disease (AD). The goal is to help the practitioner develop a senior care program incorporating behavioral screening to aid in both the recognition of behavior changes consistent with CDS and the implementation of appropriate treatment strategies.

BRAIN AGING AND COGNITIVE DYSFUNCTION

Most mammals show age-related neuropathologic changes. In humans, the most common neurodegenerative disorder is AD, which progressively impairs cognition, behavior, and quality of life. It is increasingly evident that humans, dogs, and cats demonstrate parallels in brain aging associated with cognitive dysfunction. In fact, the aged dog and, to a lesser extent, the aged cat are spontaneous models of AD and therefore can play a valuable role in testing putative AD therapeutics. Conversely, the knowledge gained from studying AD is highly relevant for understanding brain aging and cognitive dysfunction in companion animals.

Lessons Learned from AD Research

Modern medicine has increased the life span across many species, which in turn has increased the incidence of neurodegenerative diseases, such as AD. In humans, AD is generally characterized by initial decline in episodic memory followed by progressive decline across multiple cognitive domains.[1] Ultimately this results in behavioral changes that impair social function and eventually results in death. Classical diagnosis of AD has relied on post-mortem confirmation of 2 hallmark pathologies: the presence of senile plaques, which consist of extracellular deposits of fibrilized amyloid-beta protein (Aβ), and neurofibrillary tangles, which consist of intracellular paired helical fragments of cytoskeletal hyperphosphorylated tau protein.[2] However, it is uncertain if either of these pathologic changes is a causal factor as several other brain changes are documented, including neuronal loss; cortical atrophy including atrophy of the hippocampus; alterations in neurochemical systems such as the cholinergic, glutaminergic, dopaminergic, and GABAergic neurotransmitter systems; and reduced neuronal and synaptic function.[3] Moreover, risk factors associated with the development of AD include genetic, metabolic, and nutritional influences, which may be equally relevant to pet aging.

Recent clinical and research criteria propose there are progressive stages of AD from the preclinical stage (ie, prior to clinical signs) to mild cognitive impairment (MCI; prodromal stage likely to proceed to AD) to a clinical diagnosis of AD based on cognitive-behavioral status.[1] It is suggested that beta-amyloid deposition may occur early in disease progression, followed by neuronal degeneration/synaptic dysfunction (measured by biomarkers of tau pathology or functional imaging). Both of these pathologies likely precede the clinically identifiable stage of MCI or AD in humans.[4] Therefore, clinical AD is now considered a late stage of disease progression, which may explain the limited clinical success of therapeutic interventions (ie, initiated too late in disease progression to improve outcome). Therefore, identification of current and/or novel biomarkers predictive of AD progression will be essential for diagnostic characterization of preclinical and prodromal AD stages and for assessing interventions aimed at prevention or reversing progression of AD. It is theorized that AD risk

factors, similar to high blood pressure or increased cholesterol in early diagnosis of cardiovascular disease, will be identified and validated, permitting early intervention and monitoring of disease progression.

Lessons Learned from Senior Dog and Cat Research

Most mammals show evidence of brain aging and consequential cognitive deficits.[5] CDS in companion animals parallels AD progression in several respects. For example, not all aged dogs and cats show behavioral signs consistent with CDS, yet subclinical alterations in cognitive function may be present, which might eventually progress to CDS.[6] Therefore, it is prudent to commence treatment early. Understanding age-related brain pathology and cognitive dysfunction is essential to fully appreciate the potential value of using biomarkers and/or cognitive status in the future diagnosis and treatment of CDS progression.

Effects of Aging on the Brains of Senior Dogs and Cats

In canine aging, frontal lobe volume decreases, ventricular size increases, and there is evidence of meningeal calcification, demyelination, increased lipofuscin, increased apoptic bodies, neuroaxonal degeneration, and reduced neurons.[7,8] In cats, age-related pathologies include neuronal loss, cerebral atrophy, widening of sulci, and increases in ventricular size.[9–11] Perivascular changes, including microhemorrhage or infarcts in periventricular vessels, are reported in senior dogs and cats, which may contribute to signs of CDS.[7,9,12–15] With increasing age, there is an increase in reactive oxygen species leading to oxidative damage in dogs and likely cats.[9,16] Increases in monoamine oxidase B activity in dogs is reported, which may increase catalysis of dopamine with subsequent increases in free radicals.[17] A decline in cholinergic tone occurs in canine aging as evidenced by hypersensitivity to anticholinergics and decreased brain muscarinic receptor density.[18] Diminished cholinergic function is also reported in cats.[10,19] Collectively, these alterations may contribute to working memory deficits or CDS, as well as alterations in motor function and REM sleep.[9,10,20–22]

In aged dogs, cats, and humans, there are similarities in deposition of Aβ in extracellular plaques and perivascular infiltrates; however, dense core plaques seen in AD are not found in dogs or cats, suggesting canine and feline plaques are less mature than those seen in AD.[9,12,14,15,23] Moreover, similar to humans, Aβ plaque load is positively correlated with cognitive impairment in dogs.[13,14,24] By contrast, cats demonstrate more diffuse Aβ plaques than either human or dog.[9,12,15,25,26] Neurofibrillary tangles are not consistently reported in either species; however, hyperphosphorylated tau is reported in brains of aged dogs and cats, which might represent pre-tangle pathology.[7,9,13,25,26]

Overall, both dogs and cats show Aβ brain deposition and pre-tangle pathology with increasing age similar to that seen in AD progression; however, these pathologies do not achieve the severity seen in AD. Nonetheless, brain Aβ deposition may prove to be relatively early predictive biomarker of CDS consistent with preclinical and/or prodromal stages of AD.[4]

Effects of Age on Cognitive Ability of Senior Dogs and Cats

In humans, cognition is composed of multiple cognitive domains that include not only learning and memory but also executive function, language, psychomotor ability, attention, and spatial abilities. In the laboratory, there are protocols to assess many of these domains in dogs and cats. Age-related and domain-specific cognitive decline

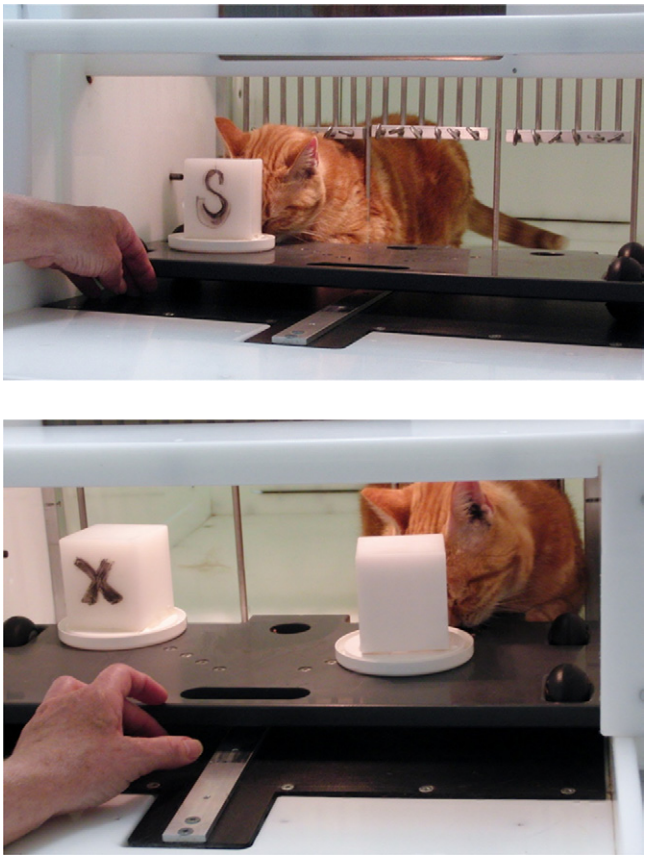

Fig. 1. The DNMP is a test of short-term visuospatial working memory.[27,30] The test consists of 2 phases. In the sample phase, the subject is required to displace an object placed over 1 of 3 possible locations on a food well (*top*); in this case the cat is required to displace block S covering food reward in the well on cat's right. The second stage (*bottom*) occurs after a delay and the subject is presented with 2 objects identical to that used in the sample phase. One object (marked with an X) is located in the same position as the sample object. The correct object is located in one of the remaining 2 positions (the nonmatch), and if the subject displaces the object, it can retrieve the food reward beneath. Initially, subjects are trained using a 5-second delay between the phases, but when the cat learns the rule that the food will always be found under the block in the nonmatch position, gradually longer delays can be introduced to assess memory.

is found in both species, but there is individual variation such that not all subjects are affected.

Using the delayed nonmatching to position (DNMP) memory task (**Fig. 1**), old dogs can be separated into 3 groups—unimpaired, impaired, and severely impaired—which may be analogous to the various stages of AD progression.[27,28] Aged dogs with DNMP impairments also demonstrate altered sleep-wake cycles, increased stereotypy, and decreased social contact with humans, which suggests a link between cognitive impairment and behavioral changes consistent with CDS.[29] Importantly, DNMP impairments can be detected as early as 6 years of age in some

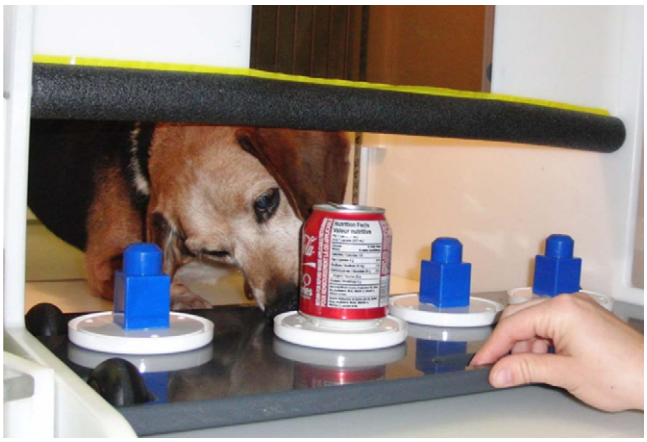

Fig. 2. In the attention task, the dog must select the correct object (covering a food reward), which is presented concurrently with either 1, 2, or 3 incorrect objects (distracters). Studies have demonstrated that performance declines and latency increases with increased distracter number, consistent with a test that assesses selective attention.[33]

dogs, which is consistent with early memory deficits in AD.[30] Also, brain amyloid deposition is reported earliest at 8 to 9 years of age.[23] Collectively, this suggests that memory impairment is an early consequence of canine aging that can precede both clinically relevant behavioral changes and amyloid deposition.

When dogs are repetitively rewarded for approaching 1 of 2 objects that differ substantially (ie, simple object discrimination learning), by contrast, no age effects on learning are evident.[31] However, if the reward contingencies are reversed after learning a simple object discrimination problem such that the dog must learn to respond to the object that previously was not rewarded in the original learning task (reversal learning), aged dogs require significantly more trials to learn to respond to the newly rewarded object than young dogs.[31,32] This impairment is analogous to the diminished executive function observed in human aging, AD, and other species.[32] On the other hand, age-related learning deficits are apparent when complex discrimination learning is assessed (eg, objects more similar or more objects), which may be related to age-related deficits in attention (**Fig. 2**).[33]

Previous studies have identified eyeblink conditioning deficits in aged cats, and a holeboard task revealed age effects on working memory, but not on spatial learning.[34,35] Age-related cognitive impairments are also seen in cats when feline adaptations of canine tests are used (see **Figs. 1** and **2**). Like dogs, cats demonstrate reversal learning and DNMP impairments with increasing age.[36] While there are insufficient data to determine age-of-onset of these impairments, reversal learning impairments were evident in 7.7- to 9-year-old cats compared to 2- to 3.8-year-old cats, suggesting that cognitive deficits precede clinical signs.[37] Similarly, while aged cats demonstrate neuropathologic brain changes similar to those reported in aged humans and dogs, the effect of neuropathologic changes on cognition have not been thoroughly investigated in the cat.[9,26] Overall, both dogs and cats demonstrate age-dependent and domain-specific cognitive decline consistent with those reported in aged humans.

Do Dogs and Cats Get AD?

Dogs and cats show both neuropathologic and cognitive changes that share many attributes of human aging and AD progression. However, late-stage AD progression is associated with impairment in most, if not all, cognitive domains. By contrast, dogs and cats do not show such extensive cognitive impairments (eg, ability to eat is retained), which suggests that the disease progression in pets is more comparable to earlier stages of AD progression. Consistent with this view, aged dogs show declining CSF levels of Aβ42, increased CSF levels of phospho-tau, and atrophy of the hippocampus, all of which are biomarkers being investigated as early diagnostic predictors of AD.[8,38,39] Future research better characterizing the longitudinal interaction among neuropathologic, cognitive, and behavioral changes in aging dogs and cats will be essential in better determining if and how CDS overlaps with AD progression.

COGNITIVE DYSFUNCTION SYNDROME

Clinical Signs of CDS

Classic signs of CDS are summarized by the acronym DISHA, which refers to **d**isorientation; alterations in **i**nteractions with owners, other pets, and the environment; **s**leep-wake cycle disturbances; **h**ousesoiling; and changes in **a**ctivity.[40] Although a decline in activity might be reported, laboratory studies suggest that increased locomotor activity and decreased immobile time are associated with greater cognitive impairment.[29,41] In addition, signs of fear, phobias, and anxiety, which are commonly reported by owners of senior pets, may be analogous to the finding of agitation and anxiety in humans with AD and might also be considered a component of CDS (**Table 1**).[42–46] Finally, memory deficits, which are some of the first recognizable signs of cognitive impairment in humans, have been identified early in the process of brain aging in both dogs and cats.[30,36] Therefore, learning or memory deficits in aged dogs and cats would also be a sign of CDS. However, these are difficult to recognize except perhaps in dogs that have been trained to a high level of performance (eg, service dogs, dogs trained for detection tasks) or by highly perceptive owners. For clinical signs of CDS, see the questionnaire in **Table 1**.

Differentiating Medical and Behavioral Problems from CDS

The determination of either a primary behavioral diagnosis or CDS must first be approached by excluding medical causes (**Table 2**). In the senior pet, this can be particularly challenging because with increasing age, there is an increased likelihood of concomitant medical conditions. Potential behavioral effects of medications must also be considered, especially those known to impact behavior. For example, steroids can increase drinking, appetite, and panting and are also associated with behavioral signs including nervousness, restlessness, irritable aggression, startling, food guarding, avoidance, and increased barking.[47] Additionally, senior pets may be less able to cope with stress, which may make them more susceptible to changes in their environment.

Thus behavioral signs in the senior pet can be due to medical or behavioral causes, cognitive dysfunction, or a combination thereof. For example, disruption of night time sleep in senior pets may be due to CDS, sensory dysfunction, or medical conditions that present with pain, polyuria, or hypertension, as well as alterations in the owner's schedule or home environment. Once problems arise, experience (ie, learning) further influences whether the behavior is likely to be repeated. In establishing a diagnosis of CDS, the clinician must be aware that the characteristic behavioral signs overlap with those of many medical and behavioral disorders.

Table 1
CDS checklist[1]

Signs: DISHAAL	Age First Noticed	Score 0–3[a]
D: Disorientation/Confusion—Awareness—Spatial orientation		
Gets stuck or cannot get around objects		
Stares blankly at walls or floor		
Decreased recognition of familiar people/pets		
Goes to wrong side of door; walks into door/walls		
Drops food/cannot find		
Decreased response to auditory or visual stimuli		
Increased reactivity to auditory or visual stimuli (barking)		
I: Interactions—Social Relationships		
Decreased interest in petting/avoids contact		
Decreased greeting behavior		
In need of constant contact, overdependent, "clingy"		
Altered relationships other pets—less social/irritable/aggressive		
Altered relationships with people—less social/irritable/aggressive		
S: Sleep–Wake Cycles; Reversed Day/Night Schedule		
Restless sleep/waking at nights		
Increased daytime sleep		
H: Housesoiling (Learning and Memory)		
Indoor elimination at sites previously trained		
Decrease/loss of signaling		
Goes outdoors, then returns indoors and eliminates		
Elimination in crate or sleeping area		
A: Activity—Increased/Repetitive		
Pacing/wanders aimlessly		
Snaps at air/licks air		
Licking owners/household objects		
Increased appetite (eats quicker or more food)		
A: Activity—Apathy/Depressed		
Decreased interest in food/treats		
Decreased exploration/activity/play		
Decreased self-care (hygiene)		
A: Anxiety		
Vocalization, restlessness/agitation		
Anxiety, fear/phobia to auditory or visual stimuli		
Anxiety, fear/phobia of places (surfaces, locations)		
Anxiety/fear of people		
Separation anxiety		
L: Learning and Memory—Work, Tasks, Commands		
Decreased ability to perform learned tasks, commands		
Decreased responsiveness to familiar commands and tricks		
Inability/slow to learn new tasks		

[a] Score: 0 = none; 1 = mild; 2 = moderate; 3 = severe.
Adapted from Landsberg GM, Hunthausen W, Ackerman L. The effects of aging on the behavior of senior pets. Handbook of behavior problems of the dog and cat. 2nd edition. Philadelphia: WB Saunders; 2003. p. 273; with permission.

Table 2
Medical causes of behavioral signs

Medical Condition/Medical Presentation	Examples of Behavioral Signs
Neurologic: central (intracranial/extracranial) particularly if affecting forebrain, limbic/temporal and hypothalamic; REM sleep disorders	Altered awareness, response to stimuli, loss of learned behaviours, housesoiling, disorientation, confusion, altered activity levels, temporal disorientation, vocalization, change in temperament (fear, anxiety), altered appetite, altered sleep cycles, interrupted sleep
Partial seizures: temporal lobe epilepsy	Repetitive behaviors, self-traumatic disorders, chomping, staring, alterations in temperament (eg, intermittent states of fear or aggression), tremors, shaking, interrupted sleep
Sensory dysfunction	Altered response to stimuli, confusion, disorientation, irritability/aggression, vocalization, house soiling, altered sleep cycles
Endocrine: feline hyperthyroidism	Irritability, aggression, urine marking, decreased or increased activity, night waking
Endocrine: canine hypothyroidism	Lethargy, decreased response to stimuli, irritability/aggression
Endocrine: hyperadrenocorticism/hypoadrenocorticism	Lethargy, house soiling, altered appetite, decreased activity, anxiety
Endocrine: insulinoma, diabetes	Altered emotional state, irritability/aggression, anxiety, lethargy, house soiling, altered appetite
Endocrine: functional ovarian and testicular tumors	Increased androgen-induced behaviors. Males: aggression, roaming, marking, sexual attraction, mounting. Females: nesting or possessive aggression of objects.
Metabolic disorders: hepatic/renal	Signs associated with organ affected: may be anxiety, irritability, aggression, altered sleep, house soiling, mental dullness, decreased activity, restlessness, increase sleep, confusion
Pain	Altered response to stimuli, decreased activity, restless/unsettled, vocalization, house soiling, aggression/irritability, self-trauma, waking at night
Peripheral neuropathy	Self-mutilation, irritability/aggression, circling, hyperesthesia
Gastrointestinal	Licking, polyphagia, pica, coprophagia, fecal house soiling, wind sucking, tongue rolling, unsettled sleep, restlessness
Urogenital	House soiling (urine), polydypsia, waking at night
Dermatologic	Psychogenic alopecia (cats), acral lick dermatitis (dogs), nail biting, hyperesthesia, other self-trauma (chewing/biting/sucking/scratching)

Abbreviation: REM, rapid eye movement.

The practitioner will need to consider physical examination findings (including neurologic, sensory, and pain assessment) along with medical and behavioral signs to select the appropriate diagnostic tests required to reveal the causes and contributing factors of a patient's signs. Identifying all influences on specific behavioral signs is

critical for both treatment selection and monitoring of behavioral and medical disorders.

Effects of Stress on Health and Mental Well-Being

Stress is an altered state of homeostasis that can be caused by physical or emotional factors that trigger psychological, behavioral, endocrine, and immune effects. Acute and chronic stress can also impact both health and behavior.[48–50] In addition, senior pets, especially those with medical or behavioral issues, may be more affected by stress and less able to adapt to change. Owners should pay particular attention to their pet‘s emotional and behavioral state as well as its appetite, sleep, and elimination to evaluate the role of stress. While enrichment can help maintain both physical and mental health, changes in the elderly pet’s household or schedule should be made slowly. Natural products or drugs may also be indicated (discussed later).

Prevalence of Behavioral Signs in Senior Pets

Spontaneously reported behavior problems

A number of studies have examined the prevalence of spontaneously reported behavioral signs in senior pets referred to behavioral specialists.[44,45] In 2 canine studies, behavioral complaints related to aggression, or fear and anxiety, were most prevalent. In a similar senior cat study, most displayed signs of marking or soiling; however, cases of aggression, vocalization, and restlessness were also serious enough to solicit referral.[50]

To further examine the distribution of problems reported by owners of senior dogs and cats, the Veterinary Information Network (VIN) database was searched for behavior problems of 50 senior dogs (aged 9–17) and 100 senior cats (aged 12–22 years). Of dogs, 62% had signs consistent with CDS, but most demonstrated anxiety, night waking, and vocalization. In the 100 feline cases reviewed, the most common complaints were related to vocalization, especially at night, and soiling. **Figs. 3** and **4** summarize the distribution of behavioral signs most commonly reported by owners of senior pets across studies.

Solicited reports of behavior problems

Since many of the most common behavioral signs in senior pets go unreported, a more proactive approach is required to establish their prevalence. In 1 study of dogs aged 11 to 16, 28% of 11-to 12-year-old dogs and 68% of 15- to 16-year-old dogs showed at least 1 sign consistent with CDS.[51] In another study of 102 dogs, 41% had alterations in at least 1 category associated with CDS and 32% had alterations in 2 categories.[52] In a more recent study, females and neutered males were significantly more affected than intact males with both prevalence and severity increasing with age, which is consistent with previous reports.[46,53] Moreover, social interactions and house training were the most impaired categories.[46]

In a recent epidemiologic study using an internet survey format of 497 dogs ranging in age from 8 to 19 years, the prevalence of CDS was 5% in 10- to 12-year-old dogs, 23.3% in dogs 12- to 14-year-old dogs, and 41% in dogs older than 14, with an overall prevalence of 14.2%. However, only 1.9% of cases had a veterinary diagnosis of CDS.[54] In 1 study of aged cats presented to veterinary clinics for routine annual care, 28% of 95 cats aged 11 to 15 and 50% of 46 cats older than 15 years were diagnosed with possible CDS.[37]

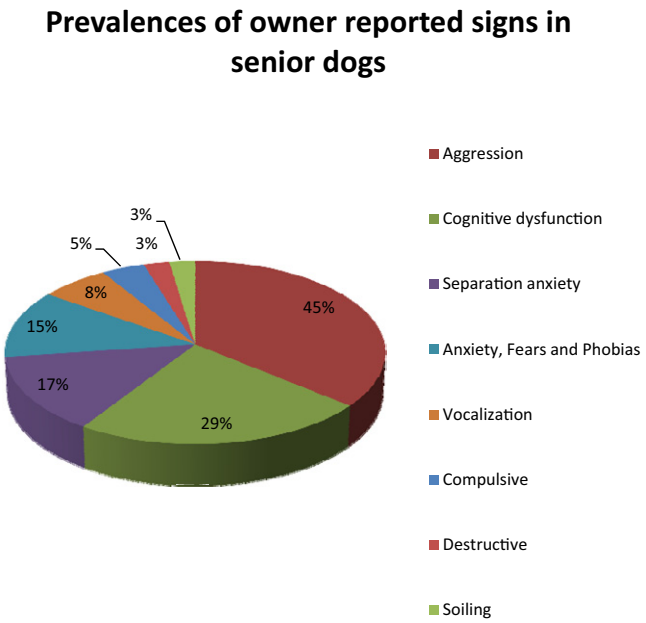

Fig. 3. Fears and phobias (includes generalized anxiety), compulsive includes repetitive and stereotypic behavior; cognitive dysfunction includes disorientation, wandering, waking and anxious at night. Behavior signs were combined from 3 studies: a Spanish study of 270 dogs older than age 7 that were presented for behavior problems, 103 dogs referred to a veterinary behaviorist, and a search of the Veterinary Information Network (VIN) of 50 dogs aged 9 to 17 years.

Importance of Client Education and Screening in the Veterinary Clinic

The data presented above indicate several important findings. First, behavioral signs related to anxiety, vocalization, night waking, soiling in cats, and aggression in dogs are more often spontaneously reported to veterinarians, which is likely related to the impact of these behaviors on the owner. Second, behavioral changes consistent with CDS are reported less frequently but are present in a significant proportion of the population. Third, the prevalence of behavioral signs consistent with CDS increases with age. Finally, because CDS is likely underdiagnosed when solicited reporting is not used, proactive monitoring and assessment of behavioral signs should be components of every veterinary visit involving senior pets. Veterinarians and their staff must inform clients of the health and welfare consequences if these problems are untreated. Handouts and web links on senior care and cognitive dysfunction syndrome can be used to further educate owners. Questionnaires are particularly effective for quick and comprehensive screening. Several are available for screening, including **Table 1**, a scoring system known as age-related cognitive and affective disorders, and a recently published 13-point data-based assessment tool.[40,54]

THERAPEUTIC OPTIONS FOR PETS WITH CDS

Behavioral Support and Environmental Enrichment in the Management of CDS

Canine studies have shown that mental stimulation is an essential component in maintaining quality of life and that continued enrichment in the form of training, play,

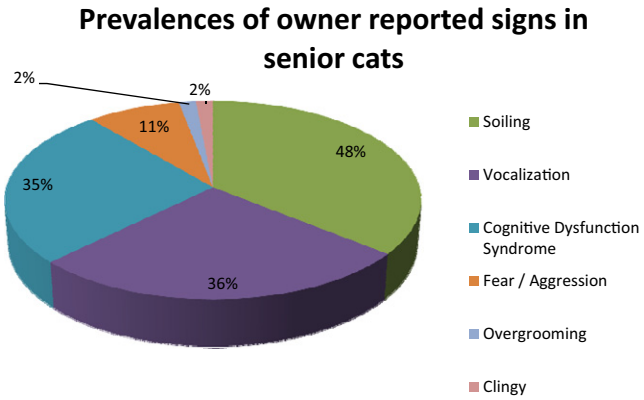

Fig. 4. Soiling includes marking, cognitive dysfunction includes disorientation, restless, wandering and night waking, and fear/aggression (includes fear and hiding). Behavior signs were combined from a VIN data search of 100 cats aged 12 to 22 years and 83 from 3 different behavior referral practices.

exercise, and novel toys can help to maintain cognitive function (ie, use it or lose it).[55] This is analogous to human studies in which increased mental activity and physical exercise have been found to delay the onset of dementia.[56,57]

Environmental enrichment can have positive effects on behavioral health and quality of life in pets and is likely to improve cognitive function.[58] Inconsistency in the management of the senior pet's environment (especially for cats) can cause stress and negatively impact health and behavioral well-being.[40,50] As sensory, motor, and cognitive function decline, new odor, tactile, and/or sound cues may help the pet better cope with its environment. Dogs with increased urine frequency may need more frequent trips outdoors or even the addition of an indoor toilet area. Ramps and physical support devices may be necessary to address mobility issues. For cats, inappropriate elimination may be improved by providing more litter boxes with lower sides and nonslip ramps.

Enrichment should focus on positive social interactions as well as new and varied opportunities for exploration, climbing, perching, hunt-and-chase games, and other stimulating ways to obtain food and treats. Food toys that require pushing, lifting, dropping, batting, pawing, or rolling to release food help older dogs and cats to remain active and alert (**Fig. 5**). By scattering favored food, treats, or catnip in different locations, pets can learn to hunt, search, and retrieve.

Maintenance of a day-night cycle by opening blinds and providing outdoor activities (where practical) to provide daylight during the day and reducing exposure to artificial light at night may be considered. Increased daytime enrichment with several quality interactive sessions, food toys, outdoor exercise (if appropriate), and a final interactive play session prior to bed may help encourage better sleep.

Drug Therapy for CDS

CDS cannot be cured at present, but deterioration may be slowed and clinical signs improved. Assuming concomitant medical and behavior problems are being controlled, various drugs (**Table 3**) may be considered to improve cognitive function or control clinical signs. For each pet, the clinician must weigh potential risks against potential benefits.

Fig. 5. A food manipulation toy, the Kong Wobbler (Kong Company, Golden, CO, USA). This toy is filled with food pieces and treats that are delivered through the opening as the dog learns to tip the toy.

Selegiline (Anipryl; Pfizer Animal Health, New York, NY, USA) is a selective and irreversible inhibitor of monoamine oxidase B.[59] It may enhance dopamine and other catecholamines in the cortex and hippocampus and has been shown both in the laboratory and clinic to improve signs consistent with CDS in dogs.[59,61] Selegiline

Table 3
Doses for drugs for behavior therapy of senior pets

	Dog	Cat
Selegiline (CDS)	0.5–1 mg/kg sid in am	0.5–1 mg/kg sid in am
Propentofylline (CDS)	2.5–5 mg/kg bid	¼ of a 50 mg tablet daily
Oxazepam[a]	0.2–1 mg/kg sid–bid	0.2–0.5 mg/kg sid–bid
Clonazepam[a]	0.1–1.0 mg/kg bid–tid	0.02–0.2 mg/kg sid–bid
Lorazepam[a]	0.025–0.2 mg/kg sid–tid	0.025–0.05 mg/kg sid–bid
Diphenhydramine[a]	2–4 mg/kg	1–4 mg/kg
Fluoxetine	1.0–2.0 mg/kg sid	0.5–1.5 mg/kg sid
Paroxetine	0.5–2 mg/kg	0.5–1.5 mg/kg
Sertraline	1–5 mg/kg sid or divided bid	0.5–1.5 mg/kg sid
Buspirone	0.5–2.0 mg/kg sid–tid	0.5–1 mg/kg bid
Trazodone	2–5 mg/kg (up to 8–10) prn–tid	Not determined
Phenobarbital	2.5–5 mg/kg bid	2.5 mg/kg bid
Memantine	0.3–1 mg/kg sid	Not determined
Gabapentin	10–30 mg/kg q 8–12 h	5–10 mg/kg q 12 h

Abbreviation: sid, once daily.

[a] Use single dosing prior to sleep or anxiety-evoking event, up to maximum daily dosing for control of ongoing anxiety.

also may be neuroprotective possibly by reducing free radical production and/or increasing enzymes that scavenge free radicals such as superoxide dismutase and catalase.[59,60] Selegiline is not licensed for use in cats but is used off label with anecdotal reports of improvement in CDS-like signs.[62] Selegiline may require 2 weeks or longer before clinical improvement is seen, should not be used concurrently with other monoamine oxidase inhibitors (eg, amitraz), and should be avoided, or used cautiously, with drugs that may enhance serotonin transmission (such as selective serotonin reuptake inhibitors, tricyclic antidepressants, buspirone, trazodone, tramadol, and dextromethorphan).

Propentofylline (Vivitonin; Merck Animal Health, Milton Keyes, UK) is licensed in some European countries for the treatment of dullness, lethargy, and depressed demeanor in old dogs. Propentofylline may increase blood flow to the heart, skeletal muscles, and brain and may have neuroprotective properties due to inhibiting the uptake of adenosine and blocking phosphodiesterase. Propentofylline has been anecdotally used in cats, but there is no clinical evidence of efficacy.

Drugs thought to enhance the noradrenergic system, such as adrafinil and modafinil, might be useful in older dogs to improve alertness and help maintain normal sleep-wake cycles (by increasing daytime exploration and activity).[63,64] However, dose and efficacy in dogs are not well established. Newer treatment strategies include the *N*-methyl-D-aspartate receptor antagonist memantine or hormone replacement therapy, but evidence is currently lacking to make appropriate suggestions for treatment.[65]

In canine and feline CDS, as well as in AD, there is evidence of cholinergic decline (see earlier). Because the elderly are particularly susceptible to anticholinergic drugs, it is prudent to consider therapies with less anticholinergic effects. Drugs and natural products that enhance cholinergic transmission might have potential benefits for improving signs of CDS, but more research is required to select appropriate drugs and doses.[66]

Nutritional and Dietary Therapy for CDS

Nutritional and dietary interventions (**Table 4**) can improve antioxidant defense thereby reducing the negative effects of free radicals. A senior diet (Canine b/d, Hills Pet Nutrition, Topeka, KS, USA) for dogs improves signs and slows the progress of cognitive decline.[67–69] The diet improved performance on a number of cognitive tasks when compared to a nonsupplemented diet as early as to 2 to 8 weeks after the onset of therapy. However, the combined effect of the supplemented diet and environmental enrichment provided the greatest benefit and, when started prior to the onset of behavioral signs, may extend cognitive health.[67]

Another strategy is a diet containing medium-chain triglycerides (MCTs), which are converted to ketone bodies by the liver. Since a decline in cerebral glucose metabolism and reduced energy metabolism are associated with cognitive decline, MCT-induced ketone bodies provide an alternate energy source that can be used by the brain. When compared to control, the diet (Purina One Vibrant Maturity 7+ Formula; Nestlé Purina PetCare, St Louis, MO, USA) significantly improved performance on several cognitive tasks.[70] Supplementation with MCTs also improves mitochondrial function, increases polyunsaturated fatty acids in the brain, and decreases amyloid precursor protein in the parietal cortex of aged dogs.[71,72] Supplementation with MCTs is also approved as a medical dietary supplement for AD patients. Cognitive diets for cats have not yet been developed.

A number of clinical trials have shown improvements in signs associated with CDS in dogs using dietary supplements containing phosphatidylserine, a membrane phospholipid.[73,74] One product (Senilife; CEVA Animal Health, Libourne, France) was

Table 4
Ingredients and doses of natural therapeutics for senior pets

	Ingredients	Dose
Senilife	Phosphatidylserine, *Gingko biloba,* vitamin B6 (pyridoxine), vitamin E, resveratrol	Dogs and cats (see label)
Activait	Phosphatidylserine, omega-3 fatty acids, vitamins E and C, L-carnitine, alpha-lipoic acid, coenzyme Q, selenium	Separate dog and cat products
Activait Cat	Note: no alpha-lipoic acid in feline version	See label
Novifit	*S*-Adenosyl-L-methionine-tosylate disulfate (SAMe)	Dog: 10–20 mg/kg sid Cat: 100 mg sid
Neutricks	Apoaequorin	Dogs: 1 tablet per 18 kg
Prescription diet b/d Canine aging and alertness	Flavonoids and carotenoids from fruits and vegetables, vitamin E, vitamin C, beta-carotene, selenium, L-carnitine, alpha-lipoic acid, omega 3 fatty acids	Dogs
Purina One Vibrant Maturity 7+ Senior	Medium chain triglycerides (from coconut oil)	Dogs
Melatonin	Endogenous-based peptide	Dogs: 3–9 mg Cats: 1.5–6 mg
Anxitane	Suntheanine	Dogs: 2.5–5 mg/kg bid Cats: 25 mg bid
Harmonease	Magnolia and phellodendron	Dogs: up to 22 kg ½ tablet daily; >22 kg 1 tablet daily Cats: N/A
Zylkene	Alpha-casozepine	Dogs: 15–30 mg/kg/d Cats: 15 mg/kg/d
Pheromones	Adaptil collar, diffuser, or spray for dogs Feliway spray or diffuser for cats	As per label
Lavender	Aromatherapy for dogs	As per label

tested in aged dogs using a cross-over design in which DNMP memory performance was improved after 60 days of treatment with Senilife.[75] Although labeled for use in cats, efficacy studies are not published.

Another product containing phosphatidylserine (Activait; Vet Plus Ltd, Lytham St. Annes, UK) demonstrated significant improvement over placebo on signs of disorientation, social interactions, and house soiling in dogs.[74] A feline version of Activait, with no alpha-lipoic acid, is also available but has not been tested in clinical trials.

Another available supplement for cognitive health (Novifit; Virbac Animal Health, Ft Worth, TX, USA) contains *S*-adenosyl-L-methionine (SAMe) tosylate, which is found in all living cells and is formed from methionine and adenosine triphosphate. SAMe may help to maintain cell membrane fluidity, receptor function, and the turnover of monoamine transmitters, as well as increase the production of the endogenous antioxidant glutathione.[76] In a recent placebo-controlled trial, greater improvement in activity and awareness was reported in the SAMe group after 8 weeks.[77] Since SAMe might increase central serotonin levels, caution should be used when combining with

other drugs that might increase serotonin. On the other hand, SAMe has been used in human patients to enhance the effects of serotonin reuptake inhibitors in the treatment of depressive disorders.[78] In laboratory aged dog and cat studies, SAMe improved measures of executive function and possibly attention.[79,80] In cats, these effects were mainly evident in the least cognitively impaired subjects, suggesting that supplementation with SAMe early in disease progression, rather than in more severely impaired subjects, should be most beneficial.[80]

Apoaequorin (Neutricks; Quincy Animal Health, Madison, WI, USA), recently released in the United States, improved learning and attention in laboratory trials compared to both placebo and selegiline.[81] Apaoequorin is a calcium buffering protein that has been postulated to provide neuroprotection in aging and consequently have positive effects on signs of brain aging.

Last, curcumin, an antioxidant, antiamyloid, and antiinflammatory compound found in the turmeric and catechin spices, is postulated to be helpful.[82]

Adjunctive Therapies for Anxiety and Night Waking

Because behavioral signs associated with anxiety and night waking are highly prevalent in senior pets and greatly impact the owner-pet bond, it is prudent for the practitioner to rapidly address them. Drugs and natural remedies that help reduce anxiety and aid in reestablishing normal sleep-wake cycles can also be of benefit in senior pets alone or in conjunction with drugs for CDS (see **Tables 3** and **4**).

Melatonin is best given to dogs 30 minutes before bedtime. Diphenhydramine, phenobarbital, or trazodone can also promote sedation. For the dog or cat that has difficulty settling at night but then sleeps well, situational use of anxiolytics can be helpful. Benzodiazepines have rapid onset, are generally short acting, and have sedative effects at the higher end of the recommended dosage range. In pets where liver function is compromised, clonazepam, lorazepam, or oxazepam is recommended because they have no active metabolites. Since pain may contribute to unsettled sleep or night waking, gabapentin can be added both as an adjunctive therapy for pain management and for its behavioral calming effects.

For senior pets with generalized anxiety, noise phobias, or separation anxiety, buspirone or selective serotonin reuptake inhibitors like fluoxetine (Reconcile; Elanco, Greenfield, IN, USA) and sertraline may be considered because of their low risk of side effects. Paroxetine and tricyclic antidepressants have varying degrees of anticholinergic effects and therefore should not be a first-choice therapeutic. However, these drugs should not be used concurrently with selegiline.

Natural compounds that may reduce anxiety and help pets settle at night include suntheanine (Anxitane; Virbac Animal Health, Ft Worth, TX, USA), honokiol and berberine extracts (Harmonease; VPL, Phoenix, AZ, USA), alpha casozepine (Zylkene; Vetoquinol Canada, Lavaltrie, PQ, Canada), pheromones (Adaptil and Feliway; CEVA Animal Health, Libourne, France), and lavender essential oils.[40]

SUMMARY

CDS is an underdiagnosed behavioral problem that affects a substantial number of aged pets. Because changes in behavior are often early indicators of medical or behavior problems in senior pets, the veterinarian faces the challenge of ruling out the influence of medical problems, sleep disturbances, anxiety, concurrent medications, and pain before a diagnosis of CDS can be made. While there are several options for treatment of CDS, many therapeutics have not been adequately tested. Moreover, early intervention is likely to be most beneficial. As we learn more about biomarkers of brain aging, objective tests for identifying pets likely to progress to CDS may be

developed. In the meantime, a proactive approach for early identification and monitoring of behavioral signs is essential for establishing a diagnosis and monitoring treatment success.

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